

Glossary

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Module 1: Classifying and Sourcing Data

Term	Definition
Continuous data	Numerical data, also called quantitative data, which represents measurements of any value within a specified range or interval. Continuous data can include angles, height, or other similar measurements.
Data	Any piece of information that you can collect, analyze, and interpret to make decisions. Data is a collection of information.
Data analysis	The process of examining, cleaning, transforming, and modeling data to help find useful information, reach conclusions, and support decision-making.
Data analytics	The application of techniques and processes to collect, analyze, and interpret data to identify trends, make predictions, and improve decision-making.
Discrete data	Quantitative data that includes integers or whole numbers, such as 1, 2, 3, their negative counterparts -1, -2, -3, or even 0. The values are separate and distinct; there can be no decimals or fractions.
Nominal data	Categories of data that have no natural numeric value, order, or ranking. Nominal data can have values such as brown eyes, blue eyes, hazel eyes, or green eyes. Analysts can group nominal data into categories, but nominal data doesn't have a meaningful order or hierarchy.
Ordinal data	Data grouped into categories that have a natural or scale-based order. Ordinal data contains values that include terms such as very likely and unlikely. Although the text items are non-numeric, these terms have value and an order of value, also known as an order of magnitude.
Primary data	Data that the researcher or someone that they directly work with collects for their own purposes, such as answering a business-related question. Researchers usually collect primary data through methods such as surveys, experiments, observations, or interviews.
Qualitative data	Also called categorical data, it is descriptive data that represents non-numeric characteristics such as appearances, experiences, opinions, behaviors, attitudes, or beliefs.
Quantitative data	Also called numerical data, it is directly measurable data that is numerical. Quantitative data is numerical, precise, and objective.
Secondary data	Data that another person, department, company, or organization collects for their own purposes, not the researcher's. Secondary data can be used by the researcher to help them reach their goals. Frequently used secondary data sources include government reports, online databases, and private businesses.
Structured data	Data, typically characterized as quantitative, that is well organized and easily read by computers.

Term	Definition
Unstructured data	Data, typically characterized as qualitative (non-numerical), that does not follow a predefined format and lacks any built-in organization or structure. Thus, it cannot be processed and analyzed by data tools and methods that analyze structured data.

Questions I have

Example: What is the difference between confidential data and private data?

My notes

Example: Confidential data is data that an organization protects from unauthorized access, such as proprietary information, source code, employee records, personally identifiable information (PII), and protected health information (PHI).

Private data is a **type** of confidential data. Specifically, it's confidential data about a **person**, such as login credentials, credit card numbers, and medical test results.

Module 2: Data Types Used in Organizations

Term	Definition
Big data	Refers to vast amounts of structured and unstructured data that grows rapidly. This data is too large and complex for conventional data processing tools to manage and analyze effectively.
Custom data	Data that is collected and tailored to a company's or individual's specific needs and goals. With custom data, companies gain a deep understanding of a particular industry, market segment, or consumer group.
External data	Information that an organization obtains from sources outside itself. External data is interchangeable with secondary data, with subtle differences. The term "external data" focuses on the source, while "secondary data" focuses on the role the data plays in analysis. External data includes public data, third-party data, syndicated data, and partnership data.
Internal data	Information that an organization generates and collects within its organization for decision-making, analysis, and operational purposes. Internal data is sometimes called primary data. Calling data "internal" generally focuses only on the source of the data relative to the organization that collected it. Calling data "primary" focuses on the role the data plays in research. Internal data includes employee data, operational data, financial data, sales data, internal communication data, marketing data, and customer data.
Interval monitoring	Collects data only at regular intervals. Interval monitoring can be an effective approach for collecting data that occurs on a regular schedule or for collecting sample data that an analyst can use to extrapolate.
Online tracking	Collects data on customers' or users' online activity. For example, online tracking helps companies determine the points online at which the customer loses interest in purchasing a product. Online tracking helps companies pinpoint these moments, learn from them, and then adjust the website to address them.
Trigger-based data collection	Involves capturing information based on specific actions or occurrences, known as triggers or events. For example, you could have an app on your mobile device that counts every notification that comes in. When a notification is pushed, the app detects the event and increments its internal count by one.

Questions I have

My notes

Module 3: Externally Sourced Data

Term	Definition
Data aggregators	Groups or businesses that package up and sell these large data sets to other businesses. For example, a retailer might use weather data from a data aggregator, such as a meteorological service, to adjust inventory and marketing strategies according to seasonal patterns.
Data attribution	The process of tracing the origin of data and verifying its accuracy and reliability. It involves identifying where the data came from, collection methods, and whether it is trustworthy and relevant to the analysis.
Data attribution statement	A statement included in a report or publication that acknowledges and credits the source of the data used in a study or analysis. The purpose of a data attribution statement is to give proper credit to the data source, establish transparency, and allow other researchers to verify and access the same data for their own studies.
Data dashboard	A visual representation of data used to track and monitor the performance of a business or organization. It typically includes charts, graphs, and other visuals to communicate key metrics and trends.
Financial data	Information that represents an entity's financial position. For example, an organization might use financial data to analyze the company's economic and financial health.
Social media data dashboards	Interactive tools that actively display and provide real-time access to data from social media platforms. These dashboards enable users to monitor, analyze, and visualize various metrics and insights related to social media activities and performance.
Social media data sets	Data collected and organized from social media platforms. These data sets actively contain a wide range of information, including user-generated content, user profiles, engagement metrics, and other relevant data points.

Questions I have

Module 4: Data Usability

Term	Definition
Arrays	A list of data items or objects associated with a fixed length, such as a numerical list of final quiz scores or names of participants in a race.
Cloud storage	Cloud storage refers to storing data on remote storage devices accessed over the internet.
Data cleansing	Data cleansing is the process of identifying and correcting incorrect, inaccurate, or incomplete data within a data set.
Data file format	Refers to the specific structure and organization of data within a file. It defines how software programs save, change, display, or analyze the data in a file.
Data normalization	Data normalization is the process of segmenting data and creating relationships between the segments. Normalization helps avoid duplication and errors and makes accessing and reading data easier.
Data standardization	Data standardization is the process of establishing consistent rules and formats for collecting and storing data to ensure it is organized, structured, and stored uniformly. Standardizing data can occur at various levels, including standardizing individual data points, data structures, and data file formats.
Data usability	Data usability refers to how easily users and systems can access and use data to drive meaningful insights.
Data storage	Data storage refers to the process of capturing, storing, and preserving digital information in a structured and organized manner for future use.
Direct-attached storage (DAS)	DAS is a type of data storage where users directly connect a storage device to a single computer or server.
Graphs	A data structure that consists of a set of data elements and a set of edges, or relationships, that connect them.
Linked lists	A chain of nodes, where each node has a piece of data and a connection to the next node.
Network Attached Storage (NAS)	Network Attached Storage (NAS) is a type of data storage accessible through a local network connected to a dedicated storage device, acting as a centralized file server. Users across the network can easily store, access, and share files, making NAS ideal for file storage, collaboration, and backup.
Queues	A linear data structure that follows the first-in-first-out (FIFO) principle. This means that you remove the first item added to the queue first.
Relational database	A relational database is now very common and uses tables, rows, and columns like a spreadsheet such as Microsoft Excel or Google Sheets.

Term	Definition
Stacks	A linear data structure that follows the last-in, first-out (LIFO) principle. This means that you remove the last item added to the stack first.
Storage area network (SAN)	A Storage Area Network (SAN) is a high-speed, dedicated network that connects servers to consolidated block-level storage, effectively making it appear as local device storage.
Trees	A data structure where the system structures data, called nodes, in a hierarchical format. A node could be an entire record or a single element, depending on the data design.

Questions I have

Module 5: Data Manipulation

Term	Definition
Data automation	Automatically analyzing and transforming data using software tools and algorithms. Automation includes cleaning, filtering, merging, and transforming data in various formats, such as CSV, Excel, or SQL.
Data manipulation	Actively modifying, transforming, or managing data to meet specific requirements, such as cleaning, filtering, merging, aggregating, or restructuring data to derive meaningful insights or perform the wanted operations on the data.
Scheduled tasks	Data manipulation and processing tasks that are scheduled to run automatically at specific times and intervals. Many data manipulation tools, such as Talend and Alteryx, can automate data manipulation tasks.

Questions I have

Module 6: Introduction to Statistics

Term	Definition
Attribute	An attribute is a measurable or descriptive characteristic of an object, event, or individual in a data set. Attributes can be quantitative, representing numerical values, or qualitative, categorizing data into groups or types.
Binomial distribution	A probability distribution in which only two outcomes are possible (success or failure, true or false, and so on), and the probability of occurrence of either of the outcomes remains the same across multiple independent trials.
Data distribution	Data distribution shows how values in a data set are spread out. It helps analysts see patterns, like which values are common and which are rare, and the overall range of the data.
Descriptive statistics	Descriptive statistics are informational statistics that summarize and simplify the important features of a data set and explain the data distribution.
Mean	The mean is a measure of central tendency that you calculate by adding all the values in a data set and dividing the total by the number of items in the set.
Measures of central tendency	Measures of central tendency describe the different ways you can use statistics to find the middle of a data set. Common measures of central tendency include the mean, median, and mode.
Median	The median refers to the middle value in an ordered data set. For an odd number of observations, it's the middle number. For an even number, it's the average of the two middle numbers.
Mode	The mode refers to the most frequently occurring number in a data set. A data set might have one mode, more than one mode, or no mode at all.
Normal distribution	A normal distribution, also known as a bell curve, is a distribution where three key measures—the mean (average), median (middle value), and mode (most frequent value)—all coincide. This alignment of the mean, median, and mode at the center is what gives the normal distribution its characteristic symmetrical bell shape.
Outliers	Outliers are very high or very low values on either side of the middle of the data set.
Poisson distribution	A probability distribution that measures the probability of an event occurring at a constant average rate within a specific period.
Range	Range is the difference between a data set's highest and lowest values.

Term	Definition
Skewed distribution	In a skewed distribution, the distribution is not symmetric around the peak of the curve. The tail on one side of the distribution is longer than the tail on the other, making it negatively skewed or positively skewed.
Standard deviation	Standard deviation measures the variation of each value in a data set from the mean.
t-distribution	A t-distribution is a type of probability distribution similar to a normal distribution. T-distribution usually is used when the sample size is small or when the population standard deviation is unknown.
Uniform distribution	A uniform distribution, also known as a rectangular distribution, has the same value for every outcome. The equal values create a symmetrical distribution, where each value in the distribution has the same probability of occurring, and the number of outcomes for each value is the same.
Variability	Variability refers to how much data points in a data set differ from each other and can also include how they differ from the mean (average) of the data.

Questions I have

Module 7: Statistical Samples to Describe a Large Data Set

Term	Definition
ANOVA test	An ANOVA test determines if there are significant differences between the means (average) of two or more groups or populations. By analyzing the variations between groups to the variations in groups, the ANOVA test determines if there are significant differences in means (averages) across the groups the researcher is comparing.
Central limit theorem (CLT)	The central limit theorem (CLT) states that if you take a lot of samples from any group of data and each sample is big enough, the averages of those samples will usually look like a bell-shaped curve or a normal distribution.
Chi-square test	A chi-square test is a statistical test used to assess whether there is a significant association between categorical variables or if the observed frequencies differ from expected frequencies in contingency tables.
Confidence interval	A confidence interval is a range of values that an estimate should fall between each time a test is performed within a certain level of confidence.
Confidence level	The confidence level is a measure of how reliable the estimates from a sample are likely to be if researchers were to repeat the study multiple times.
Hypothesis	A hypothesis is a specific, testable proposition or prediction about a population or the relationship between two or more variables. It is formulated based on observations, existing knowledge, or theory, and it is used as the basis for further investigation or experimentation.
Hypothesis testing	Hypothesis testing is a systematic statistical process used to evaluate the relationships between variables or compare characteristics across different populations. It involves the formulation of two opposing hypotheses: the null hypothesis, which assumes that there is no effect or difference between the groups or variables being studied, and the alternative hypothesis, which proposes that there is a meaningful effect or difference.
Inferential statistics	Inferential statistics is a branch of statistics that deals with making predictions or drawing conclusions about a population based on a sample.
Margin of error	Margin of error is the maximum amount, as a percentage, by which the estimate could vary from the true population.
Nonparametric test	A nonparametric test is a type of statistical test that does not assume a specific distribution for the data, making it useful for analyzing data that does not fit normal distribution assumptions.

Term	Definition
Parameters	Parameters are summary measures that describe the entire population and are typically estimated using statistics from samples. Parameters are used to infer characteristics about the entire population. For example, if the average height from several classroom samples is used to estimate the mean (average) height of all students in a school, that estimate is a parameter.
Parametric test	A parametric test focuses on parameters such as the mean or standard distribution of the population data. Parametric tests aim to determine whether there are significant differences or relationships between groups or variables.
Population	A population refers to the entire set of subjects or items of interest from which researchers and analysts want to gather data. This set might include people, animals, plants, manufactured goods, or other entities, depending on the research question. A population represents the total group that analysts want to draw conclusions about.
Sample means	Sample means refers to the average value calculated from a sample.
Sampling frame	A sampling frame is a list of all the members of a data set from which the analyst draws a sample. A sampling frame can be as large as an entire data set or a subset of a data set. The sampling frame is whatever set of data (the frame) from which the analyst draws the sample.
Standard error (SE)	The standard error (SE) is an average amount that a statistic will differ from the true value researchers want to estimate.
Standard error of the mean (SEM)	The standard error of the mean (SEM) is the standard deviation of the distribution of the sample means taken from a population. It helps analysts understand how much the measure of the sample varies.
Statistics	Statistics are summary measures calculated from the characteristics of a sample. Statistics help researchers understand the sample.
T-test	A t-test is a statistical test used to compare the means of two groups, determining if they are significantly different from each other, especially when the sample size is small and the data follows a normal distribution.
Z-test	A z-test is a statistical test used to determine whether there is a significant difference between sample and population means or between two sample means, assuming the data follows a normal distribution and the sample size is large.

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Glossary

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Questions I have

My notes

Module 8: Statistics in Decision Making and Risk Assessments

Term	Definition
Confidence level	The confidence level is a measure that tells researchers how reliable estimates from a sample are likely to be if they were to repeat a study multiple times. It represents the probability that the method of estimating values, like an average, would give a result close to the average of the entire population.
Descriptive modeling	Descriptive modeling is a way to use data to show and explain what has happened. This method helps researchers see patterns, trends, and connections in data.
Hard clustering	Clustering identifies and uses the inherent similarities in data points to form groups. Hard clustering assigns each data point to a specific cluster so the resulting data has clear groups with no ambiguity.
Machine learning	Machine learning is an automated type of predictive modeling, a process of using historical and current data to anticipate what future data may show, that uses data and algorithms to imitate how humans learn, gradually improving its accuracy.
Null hypothesis	In statistics, all tests include an assumption that there is no change from the threshold. This assumption is called the null hypothesis (often denoted as H_0).
One-tailed test	A one-tailed test is a hypothesis test that tests for a difference in only one direction.
P-value	P-value determines the likelihood that the results of an experiment or study occurred by chance. The p-value quantifies the probability of obtaining the observed data or more extreme results, assuming the null hypothesis is true.
Population	A population is the group the analyst wants to study, such as all the people in a city or all the computers on a network.
Predictive modeling	Predictive modeling is a process of using historical and current data to anticipate what future data may show.
Probability quotient	Scientists and statisticians measure probabilities as decimal values between 0 and 1, and the measure is called a probability quotient.
Sample	A sample is a small portion of the population that might or might not represent the population.
Significance level	The significance level, also known as the alpha level or α -level, is a threshold set by researchers to determine the strength of evidence required to reject the null hypothesis.
Soft clustering	Soft clustering assigns a probability to each data point for belonging to various clusters, allowing for more flexibility in how the researcher chooses to interpret the data.

Term	Definition
Threshold	A threshold is what other things are compared to in order to find differences. When conducting many tests, analysts compare the result with a threshold (also called a benchmark). That threshold provides the criteria the analyst uses to decide if the result passes or fails the test or indicates no change.
Two-tailed test	A two-tailed test is a hypothesis test that tests for a difference in two directions.
Type I error	A type I error, also known as a false positive, occurs when a researcher, based on data from a test or model, incorrectly rejects a true null hypothesis.
Type II error	A type II error, also known as a false negative, occurs when a researcher, based on a test or model, incorrectly interprets that the data from a study indicates a negative result or classification, even though the true condition is positive.
Statistical significance	Statistical significance is a way to determine whether the results of an experiment or study are likely to have occurred by chance or if they are meaningful.

Questions I have

Module 9: The Data Analysis Process

Term	Definition
Data analysis	Data analysis is the process of examining collected facts from data to discover information and make informed decisions. The data analysis process involves identifying, manipulating, transforming, and modeling data to uncover meaningful information.
Descriptive modeling	Descriptive modeling involves summarizing and visualizing the main features of a data set. Summary statistics (mean, median, mode), data visualization (histograms, scatter plots), and exploratory data analysis (EDA) are examples of this type of modeling.
Inferential statistics	Inferential statistics involves making inferences about a population based on a sample of data to draw conclusions and generalize about a larger population from a smaller sample. Examples include hypothesis testing, confidence intervals, and p-values.
Predictive modeling	Predictive modeling uses historical data to make informed predictions about future events or behaviors based on patterns analysts identify in the data. Some examples of this type of modeling are regression analysis, time series forecasting, and machine learning algorithms.

Questions I have

Module 10: Getting Deeper into the Data

Term	Definition
Actionable analysis	Actionable analysis refers to the process of using data insights to create specific tasks and plans that can drive change for an organization.
Data aggregation	Data aggregation is a technique analysts use to group data by one or more variables. For example, you might aggregate sales data by region to calculate average daily sales (AVG) for each region. This can help you identify trends and patterns in your data.
Data filtering	Data filtering is a technique to selectively display specific rows, columns, or fields from a data set based on specific criteria. For example, you can filter data to select only the customers who placed an order in January and their order date.
Data manipulation	Data manipulation is the process of modifying the structure and organization of the data, such as combining different datasets and altering the way data is aggregated or summarized to make it more useful or easier to analyze. The two most common ways to manipulate data are using a join or a union.
Data summarization	Data summarization involves creating new values by calculating value totals. For example, you could calculate the total revenue generated by individual salespeople on your staff and create sales metrics showing the total sales for different time periods.
Join	A join is an operation that combines rows from two or more tables based on a related column between them. The purpose of a join is to retrieve and consolidate data from multiple tables into a single result set.
Key hypotheses	Key hypotheses are well-defined expectations or predictions that you create based on prior knowledge, observations, or trends within your data. These hypotheses state specific outcomes or relationships that you expect to observe.
Overfitting	Overfitting occurs when an analyst designs an analysis too closely around a specific subset of data. This can lead them to conclusions that might not hold true across the population. This often results in models or conclusions that are overly complex, and that can't be applied to other data.
Sampling bias	Sampling bias occurs when the subset of data an analyst has chosen fails to accurately represent the overall characteristics of the population from which they gathered the data. This can inadvertently lead to skewed results and misinformed conclusions.

Term	Definition
Scope	The scope of a data analysis project sets the boundaries and parameters within which you will conduct your analysis. It defines which data sources, variables, and time periods are included in the analysis.
Selection bias	Subset analysis can be vulnerable to selection bias, which occurs if an analyst selects the subset based on assumptions that might not be supported by the data, leading to skewed results.
Structured Query Language (SQL)	SQL is a standardized programming language used for managing and manipulating relational databases. It allows users to query, update, and manage data and define and control access to that data.
Subset analysis	Subset analysis is the process of taking a big data set and dividing it into smaller parts based on certain rules. This helps researchers concentrate on the important information for their specific question or idea.
Union	A union is a technique used to combine two or more tables or data sets that have the same structure into a single table or data set. Unions are ideal for grouping similar data from multiple tables with the same structure, such as combining monthly sales data into a single annual data set.

Questions I have

Module 11: Communicating the Data Story

Term	Definition
Bar charts	Bar charts show data in a way that's easy to compare. They work well for listing categories in order from highest to lowest, showing how things relate to each other, or showing changes over time. Vertical bar charts are good for comparing things like amounts or numbers. Horizontal bar charts are good for long names of categories or for a lot of displayed items.
Dot charts or scatter charts	Dot and scatter charts show correlation within a data set. In a scatter chart, each point represents a pair of values, with one value plotted along the x-axis and the other along the y-axis. This can help viewers understand if a correlation between the axes exists.
Line charts	Use line charts to exhibit time and frequency on a continuum. For example, a line chart can show trends, such as sales trends over time.
Pie charts	A pie chart shows the proportions of different items in "slices," like a pizza or pie, to help visualize how each item fits into the whole. For a small number of items, pie charts illustrate how each slice represents a part of the whole category.
Stacked bar charts	Stacked bar charts are helpful for showing relative proportions between items, time, or frequency.
Waterfall charts	Waterfall charts show relative proportions or rankings. Waterfall charts are used to show how a starting value changes due to other amounts being added or taken away. These charts are helpful for seeing step-by-step how something gets to a final number.

Questions I have

Module 12: The Importance of Data Visualization

Term	Definition
Access level	The type of permission a user has to access and modify to a particular resource or system. It is a way of controlling and managing who can access and manipulate certain resources in a system or network.
Data dashboard	A tool that presents information using charts, graphs, and tables that helps users see trends, check performance, and make decisions.
Data visualization	The graphical representation of information. Data visualizations have labels and descriptions to guide the audience to understand what they are seeing and what to focus on.
Dynamic dashboards	Advanced tools that allow users to interact with real-time data through flexible, user-friendly interfaces.
Prototype	An early model or sample of a product or idea that is made to test how well it works and if it can actually be made. Data visualizations often go through prototype versions before the final version is created.

Questions I have

My notes

Module 13: Emerging Technologies and the Future of Data Analytics

Term	Definition
Artificial intelligence (AI)	Artificial intelligence (AI) is the science and engineering of building machines and computer programs that perform operations and actions that involve complex reasoning. AI uses computer science and large, complex data sets to help people solve problems in ways that are like human reasoning.
Augmented reality (AR)	Augmented reality (AR) is a technology that overlays computer-generated information onto an individual's view of the real world. AR works by using specific hardware and software to display digital content about the physical world. Devices such as AR glasses or even a smartphone with special software capture a live view of the physical world.
Blockchain	Blockchain is a type of shared database (called a ledger) that records transactions and tracks assets in a network. An asset can be anything of value, such as money, property, or data. blockchain works by linking together blocks of data like a chain. Each block contains a special code that makes it connect perfectly to the previous block, a timestamp showing when it was added, and details about transactions.
Emerging technology	Emerging technology refers to new and innovative products or services that are currently under development or in the early stages of deployment. These technologies have the potential to significantly impact both individuals and businesses, often involving breakthroughs in science or engineering.
Generative AI	Generative AI is a subset of AI that focuses on computer models that generate unique content, such as text, audio, images, and data based on existing data. Generative AI involves training algorithms to produce original and meaningful outputs that resemble human-created content. Other subsets of AI include machine learning and deep learning.

Questions I have

